Illustration 2

Complex	$[Ni(CN)_4]^{2-}$
Central metal atom/ion and its outer electronic configuration	Ni^{2+}: $3d^8$, $4s^0$
Outer orbitals of metal atom/ion	$3d^8$: ↑↓ ↑↓ ↑↓ ↑ ↑ ; $4s^2$: ↑↓ ; 4p: ⬜ ⬜ ⬜
Nature of ligand	CN^- Strong field ligand causes the pairing of 3d electrons in the metal
Outer orbitals of metal atom/ion in presence of ligands	$3d^8$: ↑↓ ↑↓ ↑↓ ↑↓ ⬜ ; $4s^0$: ⬜ ; $4p^0$: ⬜ ⬜ ⬜
Hybridisation	Coordination number - 4 Hybridsation - dsp2
Hybridised orbitals of the metal atom in the complex	$3d^8$: ↑↓ ↑↓ ↑↓ ↑↓ ; dsp^2 Hybridised orbitals: ↑↓ ↑↓ ↑↓ ↑↓ ; $4p_z^0$: ⬜
Geometry	Square planar
Magnetic property	No. of unparied electrons = 0; Hence diamagnetic
Magnetic moment (Using spin only formula)	$\mu_s = \sqrt{n(n+2)} = 0$

Illustration 3

Complex	**$[Fe(CN)6]^{3-}$**
Central metal atom/ion and its outer electronic configuration	Fe^{3+}: $3d^5$, $4s^0$
Outer orbitals of metal atom/ion	$3d^5$: ↑ ↑ ↑ ↑ ↑ ; $4s^0$: ⬜ ; $4p^0$: ⬜ ⬜ ⬜
Nature of ligand	CN^- Strong field ligand causes the pairing of 3d electrons in the metal

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